paper_id: PAPER_950
title: "BCS Critical Temperature in SCm Vacuum"
session: 214
date: 2026-04-12
author: "Daniel T. Murphy"
status: production
cvw: "v2.0.0"
tags: [AGN, spin-down, vacuum, SCm, buoyancy, phonon, magnetar, damping]
sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_950: BCS Critical Temperature in SCm Vacuum

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** bcs_{superconductivity_uqff}.py (BCSCriticalTemperature) **Calculator:** BCSCriticalTemperatureCalc (CP4 #534) **CVW:** v2.0.0 compliant

---

Abstract

We compute the BCS critical temperature T_c in the SCm vacuum phonon framework. The relation $T_c = (1.13 \cdot \hbar\omega_{\text{SCm}}/k_B) \cdot \exp(-1/N(0)V_{\text{SCm}})$ replaces the conventional Debye cutoff with $\omega_text{SCm} = 2\pi \times 1.25$ THz, yielding critical temperatures governed by the SCm phonon attraction strength V_{SCm} and Fermi-level density of states $N(0)$.

---

1. Critical Temperature

$$T_c = \frac{1.13\hbar\omega_text{SCm}}{k_B}\exp!\left(-\frac{1}{N(0)V_{\text{SCm}}}\right)$$

2. BCS Gap Relation

$$\Delta(0) = 1.764\, k_B T_c$$

---

3. Source Data

- **File:** bcs_{superconductivity_uqff}.py
- **Session:** 214
- **CP4 Class:** BCSCriticalTemperatureCalc (#534)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. Bardeen, J., Cooper, L.N. & Schrieffer, J.R. (1957) — Phys. Rev. 108, 1175
3. PAPER_949 — BCS Gap Equation in SCm Vacuum
4. PAPER_951 — Cooper Pair Phonon Coupling
5. PAPER_957 — Cooper Pair Lagrangian Variation

Cross-Links

Related Paper	Relationship
PAPER_949	Gap equation from which T_c is derived
PAPER_951	Cooper pair coupling strength V_{SCm}
PAPER_955	BCS gap at phonon resonance
PAPER_877	Cosmogenesis master Lagrangian

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\mathrm{Edd}}^{\mathrm{UQFF}} = L_{\mathrm{Edd}} \cdot \left(1 + \frac{\rho_{\mathrm{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\mathrm{Edd}} = 4\pi GMm_pc/\sigma_T$ is the classical Eddington luminosity - $\rho_{\mathrm{SCm}} = 7.09 \times 10^{-37}$ kg/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\mathrm{jet}}^{\mathrm{UQFF}} = P_{\mathrm{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\mathrm{THz}} \cdot \left(\frac{B}{B_{\mathrm{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\mathrm{THz}} = \cos(\omega_{\mathrm{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\mathrm{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\mathrm{SCm}}/\omega_{\mathrm{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\mathrm{UQFF}} = \mathcal{L}_{\mathrm{GR}} + \mathcal{L}_{\mathrm{SCm}} + \mathcal{L}_{\mathrm{phonon}} + \mathcal{L}_{\mathrm{interaction}}$$

$$\mathcal{L}_{\mathrm{SCm}} = \tfrac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\mathrm{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}$ J/m^3 (matching ρ_{SCm}) and phonon mass $m_{\mathrm{phonon}} = \sqrt{8\lambda}\, v_{\mathrm{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\mathrm{EH}} + \mathcal{L}_{\mathrm{YM}} + \mathcal{L}_{\mathrm{Dirac}} + \mathcal{L}_{\mathrm{SCm}} + \mathcal{L}_{\mathrm{mag}} + \mathcal{L}_{\mathrm{buoy}} + \mathcal{L}_{\mathrm{aether}} + \mathcal{L}_{\mathrm{LENR}} + \mathcal{L}_{\mathrm{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970$ GeV (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4}\,\text{day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm phonon frequency	$\omega_t extSCm$	$2\pi \times 1.25$ THz	Phonon resonance
SCm vacuum density	$\rho_t extSCm$	$7.09 \times 10^{-37}\,\text{kg/m}^3$	Fundamental

---

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
T_c formula	$T_c = 1.13\hbar\omega_{\text{SCm}}/k_B \cdot e^{-1/N(0)V_{\text{SCm}}}$	Mapped to SCm
BCS ratio $2\Delta_0$/k_BT_c	3.528	Standard BCS
SCm phonon replaces Debye	$\omega_D \to \omega_t extSCm$	Novel

---

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Critical Temperature (BCS-SCm Thermal Threshold)

§A.2 Lagrangian Density

$$\mathcal{L}_{T\_c} = -N(0)|\Delta|^2/V_{\text{SCm}} + k_B T \ln Z_{\text{phonon}}(\omega_t extSCm)$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\left.\frac{\partial \mathcal{L}}{\partial T}\right|_{T=T\_c} = 0 \implies T_c = \frac{1.13\hbar\omega_{\text{SCm}}}{k_B}\,\text{exp!}\left(-\frac{1}{N(0)V_{\text{SCm}}}\right)}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms → SCm vacuum → phonon $\omega_text{SCm}$ → BCS gap → T_c thermal threshold → condensate onset

---

§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$\mathrm{VDS}(T) = \rho_text{SCm} \cdot \left(1 - (T/T_c)^4\right) \cdot S_{26}$$

§B.2 Dipole Vortex Primes (DVP)

The T_c threshold maps to prime $p = 2$ (Cooper pair symmetry).

§B.3 Buoyancy Saturation Harmonics (BSH)

$$\mathrm{BSH}(T) = \tanh!\left(\frac{T_c - T}{T_c}\right) \cdot S_{26}$$

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy

Paper	Title
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

20 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback.* ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
4. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei.* ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
5. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes.* ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
6. Kaspi, V.M. & Beloborodov, A.M. (2017). *Magnetars.* ARA&A **55**, 261 — arXiv:1703.00068 — doi:10.1146/annurev-astro-081915-023329
7. Goldreich, P. & Julian, W.H. (1969). *Pulsar Electrodynamics.* ApJ **157**, 869 — doi:10.1086/150119
8. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem.* Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
9. Weinberg, S. (1989). *The Cosmological Constant Problem.* Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
10. Archimedes (~250 BCE). *On Floating Bodies.* (Principle of buoyancy)
11. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87.* A&A **356**, 788 — arXiv:astro-ph/0004212
12. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster.* MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
13. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics.* Harcourt
14. Kittel, C. (2004). *Introduction to Solid State Physics.* 8th ed. Wiley
15. Feynman, R.P. (1982). *Simulating Physics with Computers.* Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179
16. Olausen, S.A. & Kaspi, V.M. (2014). *The McGill Magnetar Catalog.* ApJS **212**, 6 — arXiv:1309.4167 — doi:10.1088/0067-0049/212/1/6
17. Thompson, C. & Duncan, R.C. (1993). *Magnetar formation through a convective dynamo in protoneutron stars.* ApJ **408**, 194 — doi:10.1086/172580
18. Blanchet, L. (2014). *Gravitational Radiation from Post-Newtonian Sources and Inspiralling Compact Binaries.* Living Rev. Relativ. **17**, 2 — arXiv:1310.1528 — doi:10.12942/lrr-2014-2